

GUYANA | EXPLORATION TO PRODUCTION

THE GUYANA SEQUENCE

Petroleum Governance, Environmental Stewardship, Natural Resource
Fund Design and Lessons for Emerging Producers

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First researched chapters and a documented source register. The full 140-160 substantive-page publication and external expert review remain pending.

Source evidence cut-off: 27 September 2026. Observed 2026 financial data are through June unless indicated.

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Executive research brief

Read alongside MANUSCRIPT_WORKING_DRAFT.md and SOURCE_REGISTER.md. The final white paper is not yet externally reviewed or complete.

Purpose

Reconstruct the institutional and commercial sequence behind Guyana's transition to a large offshore petroleum producer. Test where its history supplies documentary evidence for another country preparing for first oil, and where forest endowment, geological scale, fiscal terms and local political institutions limit direct transfer. The intended audience is senior officials, new petroleum regulators, central banks, finance ministries, environmental agencies, civil-society researchers and offshore investors.

Verified sequence

Guyana has decades of exploration history. Liza was announced as commercially discovered in May 2015. The Stabroek petroleum agreement was entered in June 2016 and the official block register shows an October 2016 effective date. The first Natural Resource Fund law received assent on 23 January 2019, before oil production commenced in December 2019. The 2021 statute replaced the first framework and 2024 legislation changed withdrawal limits. The fourth operating FPSO, Yellowtail, began production in August 2025. Four FPSOs were operating during the first half of 2026. This chronology separates pre-oil institutional decisions from subsequent modifications. [S01-S05, S11]

September 2026 evidence snapshot

The September 2026 Ministry of Finance mid-year report, describing activity through June, records 162 crude lifts, of which 21 represented government profit oil. During the half-year, government received USD 1,778.6 million in profit-oil proceeds and USD 218.4 million in royalties. The Natural Resource Fund stood at USD 4.2942 billion on June 30 after USD 1.02 billion of first-half withdrawals. The ministry revised its projected full-year 2026 petroleum deposits to USD 6.4976 billion. The latter number is a projection rather than revenue already received. Government estimated overall real GDP growth of 33.3% and non-oil growth of 10.1% for the first half; GDP estimates do not directly show changes in household poverty. [S07, S10]

The distinction most useful to a new producer

The 2016 Stabroek PSA and Guyana's later model PSA do not have the same fiscal terms. For 2016 Stabroek, the published administrative summary describes a 2% royalty, a cost-recovery ceiling of 75% and equal sharing of profit oil. Under the government revenue authority's simplified binding-cap example, 12.5% profit oil plus 2% royalty yields 14.5% of gross revenue. The figure changes as recoverable costs change. The later model for new licensing announced a 10% royalty, 65% maximum cost recovery and 10% corporate tax. The two regimes must be analysed separately, along with the information available at the time each was negotiated. [S02, S14-S15]

Institutional subjects for detailed scrutiny

1. Early-law timing. The NRF predates first oil, but the institution changed over time. Assess legislation, actual reporting, parliamentary authorization, investment stewardship and expense outcomes individually.
2. Fiscal and contract administration. Petroleum volumes, government lifts, sales, cost-bank reductions, royalties, cash deposits and budget transfers are different measures. Cost-oil dispute resolution and audit resources matter alongside contract percentages.
3. Environmental assurance. Project-specific EIAs and permits, monitoring, emergency assets, enforceable insurance, operator liability and cross-border response are separate protections. The 2025 pollution act was enacted, but the 2026 mid-year report described implementation regulations and commencement as ongoing work. The May 2026 Court of Appeal decision on financial guarantees requires analysis of the full judgment and underlying instruments.
4. Economic absorption. A high fiscal balance does not establish that roads, hospitals, schools, housing and flood defenses meet service and maintenance targets. IMF staff in July 2026 called for long-run fiscal guideposts, spending-outcome monitoring, procurement compliance and stronger cost-oil audits.
5. Social and community outcomes. Gross domestic supplier spend is not local value added. Old poverty-survey results cannot demonstrate current welfare. Indigenous village allocations require programme-level and locally grounded verification.
6. Transparency and institutional participation. The EITI Board launched Guyana's delayed second validation in May 2026 and recorded concerns about confidence in the independence of civil-society representative selection. The final paper will capture independent stakeholder responses and the final validation decision if published by the cut-off.
7. Parallel climate systems. Guyana's 2009 LCDS and Norway partnership preceded oil. Carbon-credit contracts, payments and village benefit-sharing require separate accounting from the NRF and domestic petroleum emissions. Forest results cannot be transferred mechanically to reef-based or low-forest island states.

Comparator lenses

Ghana's 2011 revenue law offers distinct stabilization and heritage funds and public publication requirements. Norway's fund documents describe transferring net state petroleum cash into a large investment fund and a long-run real-return-linked fiscal guideline. Timor-Leste's 2005 Petroleum Fund provides another small-state legal design. Suriname's GranMorgu was still under construction in September 2026, with first oil planned for 2028, making the country a relevant pre-production comparison rather than a second operating case. No design is treated as an automatic policy choice for another country. [S24-S27]

Implementation workbook for a newly discovering jurisdiction

Exploration: Record basin geology and uncertainty, licensing authority and contract disclosure, ownership, environmental baseline, marine response capabilities, available technical expertise and prospective contingent liabilities.

Commercial discovery: Build independent resource and development assessment, model fiscal terms on common assumptions, identify costs and the risk allocation actually negotiated, obtain project-specific environmental assessment, map nearby fisheries and communities, establish audit rights and publish authorized agreements subject to legitimate protections.

Before first oil: Test metering and valuation, recoverable-cost records, royalty invoicing, financial custody, parliamentary procedures, periodic public reporting, spill drills, grievance channels and beneficial-ownership records. Determine budget contingencies without booking forecast oil prices as guaranteed receipts.

First two years of production: Publish reconciled lifts, realized prices, royalty and profit-oil cash, fund balances, parliamentary withdrawals, government project execution and all material environmental incidents. Check whether procurement and supplier development result in verified domestic value.

Scaling: Stress-test price changes, new project cost-bank additions, maintenance obligations, delayed infrastructure and major spill financial exposure. Compare national growth with independently sourced real household outcomes.

Publication acceptance conditions

Complete the full multi-chapter manuscript and original, reproducible exhibits. Independently reconcile NRF period accounts, obtain the Court of Appeal judgment and original insurance and permitting instruments, resolve the legal commencement status of the 2025 spill act, incorporate final EITI 2026 status, document Indigenous stakeholder evidence, check page-specific citations and have relevant experts challenge fiscal, legal, environmental and social conclusions. Count only substantive PDF pages, inspect charts and citations, and describe independent external review as pending unless completed.

Primary citations: S01-S27 in SOURCE_REGISTER.md. The manuscript is an initial researched deliverable, not a 140-160-page finished publication.

2026 observation snapshot

2015	May Liza commercial discovery	[S01]
Jan 2019	NRF statute assented before first oil	[S04]
Dec 2019	Liza Phase 1 first oil	[S01]
Aug 2025	Yellowtail became fourth producing FPSO	[S11]
Jun 2026	End-June NRF cash: USD 4.2942 billion	[S07]
Jun 2026	H1 authorized withdrawals: USD 1.02 billion	[S07]

Dates refer to distinct events. Production capacity is not actual annual output. Fund cash is not contracted or forecast future revenue.

Worked fiscal exhibit: simplified 2016 Stabroek waterfall

Using the Guyana Revenue Authority's illustrative gross-revenue convention, royalty is separate and borne by the contractor group. A 75% cost-recovery cap is a maximum, not a permanent claim on every barrel. Values below are illustrative units out of 100, not observed 2026 receipts. [S02, S15]

Recovery used	State profit oil	State royalty	Combined state units
75% (cap binding)	12.5	2.0	14.5
50% (illustrative)	25.0	2.0	27.0
20% (illustrative)	40.0	2.0	42.0

Substantive manuscript

Abstract

Guyana's offshore petroleum expansion offers an unusually compressed record of resource discovery, contractual negotiation, institutional preparation, sovereign revenue management and rapid commercial production. Liza was announced as a commercial discovery in 2015. The Natural Resource Fund was established by legislation in January 2019 before the first offshore crude was produced in December of the same year. Four floating production, storage and offloading facilities were operating by the first half of 2026. At June 2026 the government reported a USD 4.2942 billion NRF cash balance after USD 1.02 billion of withdrawals in the first half. These observations are meaningful, but on their own they establish neither an optimal petroleum agreement nor durable welfare gains. [S01, S04, S07]

This research reconstructs events in the order they happened. It distinguishes pre-existing foundations, policy instruments established before first oil, later reforms and results still under evaluation. It examines revenue accounting, the 2016 Stabroek production sharing agreement, environmental impact assessments, oil-spill legislation, public financial management, labour and domestic suppliers, forest protection and climate finance. It then maps administrative and technical decision points for another prospective producer. The paper treats a new island state differently from a forested continental state, and a non-commercial discovery differently from an operating offshore field.

The evidence is mixed. Formal fiscal and reporting institutions preceded first oil. Government and operator data document rapid expansion and growing petroleum revenues. IMF staff in July 2026 also called for stronger oil-sector audit capacity, better procurement and public-enterprise oversight, more complete household statistics and a firmer long-run fiscal guide. The EITI Board opened a delayed 2026 validation while stating concern about the independence and stakeholder confidence of the civil-society selection process. The legal framework for pollution preparedness expanded with a 2025 act, but the June 2026 government report described regulations and commencement as work still underway. These are central facts rather than footnotes. [S07, S08, S16, S19]

A transferable case is not a list of admired policies. It is a documented sequence of institutional choices, conditions required for operation, observable results, alternative interpretations and identified points of failure.

Method and definitions

The unit of analysis is Guyana's petroleum and related national governance system from exploratory activity to 27 September 2026. The offshore Stabroek project provides the production case, but the ecological and development analysis covers Guyana as a whole. The temporal cut-off is essential: September 2026 projections about year-end output and revenue are not presented as observations.

Five evidence categories are kept apart. Legal evidence consists of texts enacted, brought into force and amended. Administrative evidence consists of official licence, permit, audit and oversight documents. Financial evidence consists of audited or dated management accounts, with

explicit cash-period boundaries. Operating evidence consists of production and commissioning records. Social and ecological evidence requires outcome data rather than expenditure or project announcements. Statements from government and companies are attributed to their authors. Independent multilateral assessments and civil-society concerns are reported with their own evidentiary limitations.

Throughout, a production capacity figure means engineering capacity, not an actual monthly mean. A crude export receipt is not public revenue. Government profit-oil lifts are not identical to sales received in the same month. A fund deposit is not a parliamentary withdrawal, and the fund's cash balance is not equivalent to the public investment portfolio. Contractual value for future carbon-credit purchases is not current cash. National GDP is not a household consumption survey. These separations prevent the apparent precision of high-level figures from concealing incompatible denominators.

1. Before discovery: institutions, geology and the timing problem

Guyana's petroleum history did not begin in 2015. Onshore work dates to the early twentieth century; offshore exploration began in the middle of that century. Seismic surveys and drilling by successive companies preceded the commercial Liza discovery. ExxonMobil's retrospective 2023 report places offshore exploration attempts in the late 1950s and describes more than forty wells drilled in the basin over decades. This history matters because a country rarely controls the exact timing of an economically viable discovery. Dry wells, capital cycles, technical interpretation and petroleum prices determine much of the interval between exploration and commercial development. [S01]

Guyana also entered the modern oil period with an existing economic and ecological agenda. Its original Low Carbon Development Strategy began in 2009. A parallel Guyana-Norway results-based forest agreement began in the same year. The government subsequently reported USD 227 million earned for forest climate services during 2009-2015. A monitoring, reporting and verification system for the forest sector developed before first oil. These were pre-petroleum policy institutions and a separate financial stream, not products of the petroleum fund. [S22]

This chronology prevents two analytical mistakes. First, attributing every later institution to the petroleum discovery erases existing public administration, regulatory experience, forest monitoring and the long history of commodity production. Second, projecting Guyana's forest-carbon results onto a small island with few forests would confuse the national resource endowment with a universally available funding instrument. The transferrable administrative lesson is the practice of mapping existing technical capacity before a commercial discovery, not the replication of Guyana's exact endowment.

In an exploration-stage country, the relevant decision documents include the geological data policy, licence register, basin environmental baseline, fiscal-risk inventory, ministerial and regulatory division of responsibilities, spill-planning jurisdiction, beneficial-ownership requirements and public disclosure rules. Each is meaningful before a first commercial find. A government may procure legal and reservoir expertise through transparent short-term appointments instead of creating a large permanent petroleum bureaucracy without proven

reserves. Whether to establish a state producer, auction acreage, negotiate directly or reserve blocks requires a jurisdiction-specific legal and fiscal assessment. The case study will therefore map decision rights and procedural safeguards rather than prescribe one commercial model.

Guyana is also climatically distinctive. Much of its population and capital stock sit on a low coastal plain subject to flooding, while large forested and hinterland regions have different infrastructure, service-delivery and land-governance needs. An offshore development changes the fiscal envelope without eliminating these geographic constraints. An independent comparison must measure whether new capital expenditure reduces lifecycle exposure to flooding, power insecurity and maintenance costs. Building a new asset is an input; a working, resilient asset delivering service over time is the intended outcome.

The period before discovery is thus analytically separate from 2015-2019. A useful exploration chronology should identify not only when a well discovered hydrocarbons but also when the state gathered geological information, published licence terms, defined ownership, set environmental baselines and strengthened its ability to challenge an operator's commercial and technical submissions. Without that baseline, governments and external observers struggle to separate inherited institutional capacity from later improvements.

2. The discovery-to-first-oil window, 2015-2019

The Liza discovery announced in May 2015 changed the commercial outlook for the offshore Stabroek block. On June 27, 2016 the government and the consortium entered a petroleum agreement, with a stated October 7, 2016 effective date in the official block register. Different public records refer to signature, agreement and effective dates. Those dates should appear separately in the final chronology. The project reached first oil in December 2019. The roughly four-year interval after discovery is the institutional preparation window studied here. [S01, S02, S03]

The 2016 Stabroek agreement reflects the conditions and bargaining positions of a frontier offshore development at the time it was signed. Any contractual comparison needs the exact agreement text, hydrocarbon uncertainty at signature, oil-price conditions, deepwater capital intensity, exploration obligations, relinquishment provisions and environmental commitments. Retrospective government revenues and later geological success do not automatically establish the value a counterparty placed on acreage before commercial certainty. Conversely, early commercial uncertainty does not remove the need to document fiscal options or expose a government to unreviewed recoverable costs.

The public Stabroek agreement has a 2% royalty, up to 75% monthly cost-recovery ceiling, and an equal division of remaining profit oil. The Guyana Revenue Authority explains that when the ceiling binds, a simplified revenue calculation gives 12.5% of gross sales as government profit oil plus a 2% royalty, or 14.5% under its stated accounting approach. That is a conditional illustration, not an unchanging government entitlement. When permitted recoverable costs use less than the cap, the profit-oil pool increases. Approved future development expenditures may replenish the cost bank. Audit treatment, allowable expenditure and valuation matter as much as headline sharing percentages. [S02, S07, S15]

The historical test of the contract therefore has at least four separate components: returns relative to the information available at signing, compliance with the signed terms, effectiveness of subsequent cost verification, and the state's ability to use later experience in different acreage awards. Public debate frequently merges these into a single judgment. This manuscript keeps them separate and will not declare a country-level winner or a contract-level overall score.

A relevant reform sequence is observable. The government later announced model terms for future offshore bidding with a 10% royalty, a 65% cost-recovery ceiling and a 10% corporate tax. Profit oil remains shared equally after cost recovery under the model. These later terms are distinct from the continuing Stabroek agreement and must never be back-applied when calculating historical or forecast revenues from the producing block. The published model documents and any final signed new licence terms require an agreement-by-agreement crosswalk. [S14]

What early fund establishment demonstrates, and what it does not

The Natural Resource Fund Act received assent on January 23, 2019, almost eleven months before first oil in December 2019. The Ministry of Finance's 2024 NRF annual report confirms the date. This is a material pre-production institutional milestone: the state adopted a statutory vehicle before recurring offshore oil cash arrived. Yet formal creation alone says little about later spending efficiency, distributional outcomes or the adequacy of subsequent withdrawal rules. Those questions require separate assessment. [S04]

The 2019 establishment also did not mean the rules remained unchanged. The 2021 law replaced the 2019 act. The withdrawal provisions changed again on February 6, 2024. A new country examining this case should distinguish statutory creation, operational readiness, independent monitoring, transparent publication, integration with annual budgets and revision procedures. A law on paper is the first link in a longer accountability chain.

3. From first oil to large-scale production, 2019-September 2026

Production commenced at Liza Phase 1 in December 2019. Liza Phase 2 followed in 2022, Payara in November 2023 and Yellowtail on August 8, 2025. Yellowtail's ONE GUYANA FPSO joined the Destiny, Unity and Prosperity vessels and took combined installed capacity past 900,000 barrels per day. ExxonMobil announced a daily production milestone of 900,000 barrels in November 2025. That one-day milestone is distinct from annual average production, and installed capacity alone never establishes actual output. [S01, S11, S12]

The national petroleum production chart extends through July 31, 2026. The Ministry of Finance's 2026 mid-year report says production from the four FPSOs exceeded 900,000 barrels per day during the first half. The same report describes the fifth Uaru project as under construction with start-up targeted for the final quarter of 2026. Future development output should remain in a prospective column until a dated commissioning announcement and verified operating record are obtained. [S07, S13]

Production is not a single variable. An operating dossier needs the count and size of producing vessels, uptime, reservoir performance, flaring events, methane monitoring, water injection,

transport logistics, crude grades, lifting dates, export prices, cost-bank balances and the share delivered to the state. The government recorded 162 crude lifts during January-June 2026, including 21 government profit-oil lifts from the four producing units. Yet receipts in the same period included payments for three lifts occurring in late 2025, while payments for some mid-2026 lifts fall later. An analytical table pairing production dates directly with cash receipts without that timing adjustment would misstate the relationship. [S07]

Guyana reported USD 1,778.6 million in government profit-oil sales receipts and USD 218.4 million in royalties during the first half of 2026. End-June NRF cash stood at USD 4,294.2 million after USD 1,020 million in withdrawals during the period and including reported interest. These figures come from the government's September 2026 publication. Independent reconciliation with Bank of Guyana monthly and quarterly releases and external audit statements is part of the final source gate. [S07]

The speed of commissioning is commercially significant but creates linked challenges. Regulatory personnel need sufficient specialist expertise and access to data as vessel numbers multiply. Shipping and aviation capacity grow along with occupational risk. Contractor spending can expand faster than qualified local supplier capacity. Power demand, roads and port assets face supply-chain constraints. Large monthly tax and petroleum receipts expand budget options while also increasing the consequences of weak procurement or poor project sequencing. None of these effects is captured by a barrels-per-day graph.

The final paper will therefore present a development-by-development table with first-oil dates, sanctioned costs as of the applicable date, FPSO investment-basis capacity, actual quarterly output, permitted emissions and flaring, environmental reporting and the regulator's approval and inspection milestones. An approved project not yet producing will have blank realized-output cells rather than copied operator forecasts.

4. The Natural Resource Fund as a fiscal institution

The conceptual purpose of a natural-resource fund is to separate volatile extractive cash inflows from the immediate authorization of public expenditure. In Guyana's case, the fund was legally established before first oil. The 2021 replacement statute specified a governance structure including a board, revised accountability arrangements and withdrawal rules. The 2024 amendment increased the statutory withdrawal envelope. A claim of good fiscal practice requires analysis of each version and its observed effects rather than treating the act as an immutable institution. [S04, S05]

The fund receives the government's defined petroleum revenue streams and investment income. Withdrawals move money to the consolidated fund under parliamentary budget authorization. The public account must allow a reader to reconstruct each stock-and-flow identity:

Opening fund cash + petroleum receipts + investment income + other recognized movements - authorized transfers = closing fund cash.

Each item needs an observation period, currency and treatment of accruals or late receipts. The initial paper uses the figures disclosed for June 2026 rather than extrapolating a final year-end

outcome. The Ministry of Finance reported USD 4.2942 billion in cash at end-June after USD 1.02 billion in first-half transfers. Its revised 2026 estimate raised full-year petroleum deposits to USD 6.4976 billion, from the much lower January budget assumption. This updated full-year estimate remains a projection, influenced in the ministry's explanation by oil prices and reduced use of the cost-recovery ceiling. [S06, S07]

The revision illustrates two fiscal hazards. First, a large increase in projected receipts can be mistaken for an equally large increase in prudent current-year spending space. The withdrawal law and parliamentary budget framework are separate from the projection. Second, a rapid decline in historical cost recovery could create short-term windfall receipts that do not recur in the same amount after a new field enters development. The 2026 mid-year report expressly notes that costs from additional projects may increase the cost bank again. [S07]

IMF staff's July 2026 statement recognized the accumulation of substantial NRF balances and Guyana's debt position, while proposing consideration of a longer-run operational fiscal guide based on the non-oil primary balance. Staff argued that unexpected gains under sustained high oil prices should be examined against the economy's ability to absorb investment and called for stronger monitoring of public spending outcomes and public-enterprise risks. These are attributed external assessments. They do not prove one single withdrawal formula suitable for all emerging producers. [S08]

Cash management is not public investment evaluation

A fund statement answers where petroleum cash is and how money moves. A public investment system answers which projects are funded, procured, built, independently inspected, maintained and used. The two systems are linked but not interchangeable. A budget line for a hospital is not a verified improvement in service quality. A construction contract is not a functional road that survives flooding. A scholarship allocation is not a measured increase in skills or labour-force productivity.

The transfer framework should therefore include a separate public investment register. Each large project would record a published business case, procurement method, contract variations, delivery date, capital and lifecycle cost, disaster exposure, expected users and independent completion evidence. Where data are absent, the final manuscript will mark an evidence gap instead of inferring success or failure. Even with high revenues, capital-intensive programmes compete for engineers, surveyors, procurement specialists, nurses, teachers, land and imported construction inputs.

For a potential new producer, the administrative sequence to document includes the statutory revenue recipient, published deposit statements, a withdrawal authorization chain, parliamentary scrutiny, an external audit schedule and the project-level outcome ledger. The precise spending-versus-saving rule is a sovereign policy choice shaped by development needs, reserve life, debt, population, climate exposure and macroeconomic conditions. This research describes alternative designs and their trade-offs without assigning a universal answer.

5. Fiscal contract oversight and the cost bank

Production sharing is often explained as a royalty followed by a cost-oil allocation and a split of the remainder. That summary is insufficient for technical oversight. A credible government revenue forecast requires a model of recoverable cost categories, timing and approval of development expenditures, production valuation, commodity prices, crude quality adjustments, operating expenses, currency and transfer-pricing controls, decommissioning treatment and dispute procedures.

Under the 2016 Stabroek framework as summarized by the Guyana Revenue Authority, the simplified case in which the 75% cap is fully used allocates 75 of each 100 normalized revenue units to cost recovery, 12.5 to government's half of remaining profit oil and 12.5 to the operator group. A separate 2-unit royalty produces the familiar 14.5-unit combined government royalty/profit share under the authority's calculation. This is an explanatory model rather than a substitute for detailed contract accounting. In other periods the ceiling need not bind. [S02, S15]

At a hypothetical 50-unit cost share with a 100-unit revenue base, a simple 50/50 remainder gives the state 25 units of profit oil plus 2 units of royalty. At a hypothetical 20-unit share, the same pedagogical formula gives 40 plus 2. Actual payments need the contract's detailed oil-volume and valuation procedures, and this illustration excludes tax treatment, timing, carried costs and cash-versus-kind settlement. No reader should interpret 27% or 42% as an observed 2026 fiscal take or a forecast.

The 2026 mid-year report reported accelerated cost recovery as the historical cost bank was reduced and projected a substantially higher number of government lifts than expected in January. The report warns that new projects enter the cost bank and that clearing historical costs does not create a permanent government claim to half of gross sales. Such cautions belong in the main text because the choice of production denominator changes the interpretation of public revenue. [S07]

Contract administration also depends on verifiable audit capacity. The July 2026 IMF mission noted ongoing efforts to resolve cost-oil audits, including arbitration, and emphasized timely resolution, specialist audit staffing and the quality of government oversight. An administrative case study must distinguish allegations made in disputes, expenditures accepted as recoverable, auditor adjustments proposed, adjustments finally collected and open proceedings. A proposed disallowance is not recovered public revenue. [S08]

The later model fiscal terms illustrate a second issue: the government's bargaining position and acceptable return vary across the discovery cycle. The announced model's 10% royalty and 65% maximum cost recovery, together with equal residual profit oil and a 10% corporate tax, differ from the 2016 block agreement. A comparison should show gross expected government receipts, investor breakeven and downside geological outcomes under common assumptions; it should not infer comparative welfare from two headline royalty rates. [S14]

6. Environmental law, liability and offshore operations

Petroleum regulation should begin with an accurate inventory of harms, affected populations, potential transboundary effects and the physical capacity to respond. For a coastal state the issues include coastal water and fisheries, tourism, biodiversity, emergency maritime aviation, ocean currents, offshore worker safety, greenhouse-gas emissions, shore-based waste and possible liability beyond national waters. Guyana's EPA operates a formal environmental authorization process for projects with environmental impacts. The existence of a permit is a necessary legal record, not proof that its terms have been independently monitored or that every risk is extinguished. [S21]

The sequence should distinguish environmental assessment before project approval, permit conditions during construction, real-time operating monitoring, incident reporting, post-incident enforcement, insurance coverage and financial recourse. It should also distinguish public access to raw monitoring results from operator-authored summaries. A complete research exhibit would list each operating FPSO, its applicable environmental permit and environmental impact assessment, permitted flaring limits, reporting frequencies, available inspections, observed incidents and documented remediation.

Guyana enacted the Oil Pollution Prevention, Preparedness, Response and Responsibility Act in 2025. It addresses oil-spill prevention, planning, national response and financial responsibility. The 2026 mid-year report, covering January-June, says implementing regulations were still being drafted and that commencement was expected by the end of 2026. At the 27 September cut-off this report does not establish that the entire statute was already operational. The final legal chapter must check the Official Gazette and commencement instruments before moving a provision from 'enacted' to 'in force'. [S19, S07]

Financial assurance illustrates why precise legal language matters. In May 2026 the Court of Appeal overturned a 2023 High Court order requiring an unlimited guarantee under an offshore permit. The government's description of the appellate decision says the EPA had lawful discretion over the amount of financial assurance and that the High Court had improperly substituted its own decision for the regulator's judgment. A contemporary independent report also distinguishes an operator's continuing liability from the amount of security furnished in advance. Neither point, alone, resolves the empirical question of whether funds available following a catastrophic release would meet cleanup, compensation and ecological restoration costs. An authenticated judgment and actual policy documents are required for final legal analysis. [S20]

An offshore assurance model has multiple layers. Direct operator resources, insurance, any parent-company guarantee, rapid-access emergency cash, contracted response assets, reciprocal regional assistance and state contingency planning differ in legal trigger, latency, currency, scope and enforceability. They should not be merged into a headline figure without examining policy exclusions, aggregate limits, arbitration, who controls claim settlement and conditions after a multi-jurisdiction spill. For island and continental states alike, cross-border current modelling and neighbours' ability to respond are material planning inputs.

A future final chapter will include a reproducible response drill template: discovery-to-notification time, mobilization time, dispersant approval procedures, mechanical recovery resources,

sensitive-habitat priority maps, fishery-closure authority, independent incident sampling, medical surveillance and claims processing. Without disclosed observed drill performance, the template remains a prospective audit instrument rather than a finding about current operational readiness. Forest conservation and offshore pollution are separate environmental systems and must be evaluated on their own evidence.

7. Forest conservation, carbon finance and the fossil-fuel coexistence problem

Guyana's 2009 LCDS predated oil production by a decade. Its results-based forest partnership with Norway covered 2009-2015 and the 2024 LCDS update reported USD 227 million earned in that period. In 2022 the country introduced an expanded LCDS 2030 and arranged a multiyear sale of ART TREES jurisdictional forest credits to Hess. The published minimum purchase commitment is USD 750 million, with specific prices and volumes by credit vintage. The policy states that 15% of associated receipts are dedicated to eligible Indigenous village-led plans, and the remaining share finances wider LCDS priorities. A contract for future delivery is distinct from credits already issued, transferred, paid for and spent. [S22, S23]

This history permits analysis of two distinct resource-finance channels: extractive revenue deposited in the NRF and ecosystem-service payments managed through LCDS financial structures. A combined national wealth view is useful for climate and budget choices, but it would be inaccurate to describe all carbon receipts as oil revenue or add contracted future carbon sales to cash in the NRF. Oil production is a fossil-fuel supply activity; forest-credit issuance depends on independently specified carbon-accounting baselines, monitoring, additionality and leakage treatment. One does not mechanically cancel the other. Any claim of national carbon neutrality requires economy-wide emissions and accounting safeguards, not a simple subtraction of a contracted offset volume from petroleum exports.

The transferable question is whether a resource-producing state maintains independently verifiable environmental accounting for different assets. In Guyana's case, the forest monitoring and results-payment experience is relevant to land-use data quality and international partnership design. Its applicability to another jurisdiction depends on ecosystem type, forest tenure, historical deforestation, Indigenous rights, corresponding adjustments and carbon buyer eligibility. A reef-based island has different biodiversity and marine accounting issues. Carbon-market prices, certification methods and buyer eligibility are also vulnerable to change.

LCDS policy statements about village allocations, sea defenses, electricity investment and community livelihoods need outcome-level tests. A national transfer proves expenditure authorization, but independent review would examine arrival in designated accounts, locally agreed development plans, procurement documentation, functioning assets and households reached. Consultation must identify which communities participated, which declined and how grievances were addressed. A strategy can carry the language of inclusion while still requiring village-level verification.

IMF staff in July 2026 highlighted Guyana's market-based forest conservation, disaster-resilience investments and energy transition, while also identifying needs for stronger official statistics and procurement. These assessments do not eliminate the analytical tension between increasing fossil

output and the global energy transition. The final analysis will report domestic operational emissions separately from combustion of exported oil, land-use emissions and carbon-credit issuance. A policy commitment and a measured atmospheric outcome belong in different columns. [S08]

8. Local value, social outcomes and accountability

Guyana's Local Content Act was passed in December 2021. The law establishes a formal route for specified Guyanese participation and reporting by petroleum operators and suppliers. The 2026 mid-year report estimates roughly USD 466.2 million in procurement across sectors prioritized for Guyanese participation during the first half of the year, while government work on possible expansion of the law's scheduled categories was still underway. The number measures gross procurement in designated sectors. It does not independently reveal domestic value added, local beneficial ownership, wage quality, profits retained, product standards or the effect on non-petroleum employers. [S05, S07]

The final local-value chapter will disaggregate contractors, service categories and ownership. It will use an explicit distinction between a registered Guyanese supplier and domestic production content. Examples include port services, welding, catering, transport, engineering, aviation, environmental monitoring and professional services. An imported piece of equipment resold by a locally registered firm generates different domestic value from an engineered product manufactured and maintained in Guyana. The public data should be audited against supplier certification and customs records where legally available, with confidentiality and commercial rights preserved.

Rapid resource expansion also changes wages and the availability of teachers, nurses, civil servants and smaller-firm engineers. Government's 2026 mid-year report discusses technical and professional skills shortages and potential execution delays. Fiscal transfers and construction can raise activity while pushing against constraints in local land, machinery and skilled labour. An evidence-led study needs salary indexes, migration measures, training completions and occupational vacancy data by region. National job announcements or scholarships alone do not establish net employment and long-run earnings effects. [S07]

Social outcomes are especially susceptible to mistaken timing. The World Bank's country overview reports an older national poverty estimate based on 2019 data and explicitly notes the absence of recent comparable poverty data in that published summary. It would be misleading to treat that figure as the current 2026 poverty rate. An updated household survey, real regional consumption, nutrition, access to safe water, school completion and hinterland service coverage offer a stronger basis for later conclusions. The final paper will identify missing indicators rather than invent a post-oil poverty trajectory. [S10]

Transparency has several layers: formal contract publication, beneficial ownership, production volumes, government receipts, cost-oil audits, independent regulation, public participation and spending outcomes. Guyana has published several petroleum documents and EITI national reports. Its December 2025 EITI country report covered the 2023 fiscal year. The time lag matters for a rapidly expanding industry. [S18]

The EITI Board's May 2026 decision launched a full validation starting May 15 following a delay and stated concern about a lack of stakeholder confidence in the independence of civil-society representatives' selection process. This is a documented disagreement about the quality of participation, not grounds for this manuscript to pronounce on the legitimacy of a political actor. The final report will present the EITI decision, views of the multi-stakeholder parties and subsequent verified developments without asserting the outcome of an unfinished evaluation. [S16, S17]

More frequent digital publication can improve timeliness but will not replace independent challenge. A public resource ledger becomes useful when readers can reconcile production, sale prices, contract obligations, royalty payments, NRF deposits and authorized budget transfers. That ledger should carry dataset versions, source URLs and correction notices. A complete digital publication should be intelligible to citizens and technically testable by researchers.

9. Five international comparator designs and limits

Ghana's Petroleum Revenue Management Act 2011 established a stabilization fund and a heritage fund, along with publication and parliamentary accountability provisions. This provides a comparator for separating short-run expenditure stabilization from long-run financial savings and for setting statutory reporting frequencies. Comparing statutes alone would not show which country delivered better outcomes: data on implementation, withdrawals, public debt and actual welfare need to be paired with each legal instrument. [S24]

Norway's Government Pension Fund Global receives the state's net petroleum cash flow and sits within a fiscal framework linked to an estimated 3% long-term real return. Its 2025 official white paper discusses a guideline for keeping normal-year withdrawals below that rate and maintaining buffers for adverse years. Norway has a different institutional history, population, capital market environment and mature stock of financial assets. A new producer cannot translate its numerical fiscal guideline without modelling local infrastructure deficits, expenditure absorption, debt and reserve uncertainty. [S25]

Timor-Leste established a Petroleum Fund by law in 2005 and amended that framework in 2011. Its early fund architecture offers a comparison for a small developing state facing large prospective petroleum receipts. The final paper will add verified 2026 data on petroleum-field production, fund withdrawals, budget reliance, and development outcomes before drawing conclusions about the performance of its particular formula. A statutory rule and its fiscal sustainability are separate propositions. [S26]

Suriname offers a regional pre-first-oil comparison. TotalEnergies' September 19, 2026 update reports continuing construction at the GranMorgu offshore development, with first oil planned for 2028. It is therefore a prospective comparison, not another operating case. The administrative question concerns which Guyanese preparations could be examined before Suriname's commissioning and which differ under its national oil company, contract terms, marine environment and institutional structure. [S27]

Small island states must be handled on their own terms. A prospective producer with no tropical forest sector cannot finance development through Guyana's precise jurisdictional forest-credit

structure. A shallow-water or nearshore field presents a different spill-response geography from deepwater Stabroek operations. Tourism and reef exposure, narrow fiscal revenue bases, hurricane risk, desalination, land scarcity and cross-border response agreements change the feasibility of proposed institutions. The resulting handbook will be a configurable set of decision records, not a claim that one nation's contract and fund law suit every geology or society.

10. Transferable administrative sequence and adverse cases

A sequence-based framework begins with the premise that sovereign choices are made under different degrees of uncertainty. During exploration, the uncertainty is commercial: hydrocarbons may never be discovered in economic quantities. After a discovery, the challenge is negotiating rights and environmental protections before irrecoverable infrastructure is commissioned. At first oil, financial accounting and incident reporting need to work reliably. During scaling, the bottleneck often shifts toward public investment capacity, cost-oil review, labour, supplier quality, maintenance and exposure to global prices.

An early-country workbook would document the following actions and evidence without deciding the jurisdiction's political choices: establish a public chronology of licence and environmental authorities; inventory all fiscal obligations and contingent liabilities; map marine and coastal ecosystem baselines; obtain independent reservoir, legal, accounting and spill expertise; publish contract versions and define justified confidentiality exceptions; describe who verifies volume and price; and simulate an offshore emergency with available regional partners. Each item has a named evidence document and an owner, rather than a generic assertion of preparedness.

The second workbook starts when a commercial discovery is established. It records the development proposal's production range, cost envelope, decommissioning plan, environmental alternatives and labour forecast. It creates a source-and-use-of-funds bridge under different oil prices and differentiates operator spending from direct government expenditure. It identifies whether fiscal rules would operate before or after first cash receipts, which body publishes fund statements and which institution investigates alleged audit anomalies. Political decisions about contract models, ownership structures or spending ratios remain open; the workbook presents their quantifiable constraints.

A first-oil readiness audit would test whether field meters, cargo valuation, royalty invoices, cost-bank records, central bank reconciliations, incident-response roles and legislative reporting function under realistic load. A regulator should not assume every new FPSO offers the same risk profile or environmental permit conditions. A simple flowchart linking the operator's field data to the revenue authority, finance ministry, NRF custodian and parliamentary appropriation creates a traceable basis for independent scrutiny.

Adverse case A: a long low-price interval

A sustained fall in realized crude prices compresses gross sales, royalties and profit oil. If economic activity depends on imported equipment, construction can also slow. A test should compare projected authorized spending against revised legally available fund transfers, fiscal buffers, project postponement clauses, debt service and protection of essential operations. The same price shock has different effects when the PSA cost cap binds and after capital costs decline. Never assume a fixed state share of gross revenue across all phases.

Adverse case B: newly approved development replenishes the cost bank

A field reaches high production and government receipts rise as historical development costs decline. The operator then proceeds with further sanctioned projects. New recoverable expenditures affect the profit-oil calculation. A model based on a single high-receipt period overstates durable public cash if it ignores this cycle. Preserve fields, vintages, cost categories and contractual cross-charging constraints separately.

Adverse case C: a major offshore spill

The study assumes a serious release during adverse sea conditions and asks which legal provisions are in force, who orders emergency action, when cash becomes available, which state pays upfront, what coverage exclusions apply, how claims enter foreign jurisdictions and how independent evidence is preserved. The hazard exercise tests institutions, not an asserted probability of catastrophe. The final paper requires independent review of actual emergency plans and contractually enforceable insurance terms.

Adverse case D: overheating without fiscal insolvency

Even with a rising financial fund balance, public construction demand may exceed available contractors, engineers and aggregate supply. Prices and wages rise, project delays accumulate and the non-oil economy loses competitiveness. The July 2026 IMF statement highlights absorption, procurement and monitoring concerns as questions for ongoing policy review. The stress test therefore tracks real input costs and completed units of service, not budgeted nominal spending alone. [S08]

Adverse case E: public confidence breaks despite disclosure

An extractive revenue portal posts thousands of data rows while representatives of affected communities question the independence of stakeholder selection or regulators' responses. Mechanical disclosure does not substitute for meaningful access to information, independent representation and a process for correcting data. The EITI Board's May 2026 statement makes the distinction concrete. The audit would trace how complaints are recorded, decisions documented and adverse findings disclosed. [S16]

11. Initial findings and publication boundaries

Five verifiable facts structure the emerging case. The first commercial discovery preceded first oil by roughly four years. A Natural Resource Fund statute was assented to before first production. Subsequent fund governance and withdrawal rules changed in 2021 and 2024. Four offshore FPSOs were producing by the first half of 2026, with growth in reported profit-oil receipts and substantial cash held in the NRF. Guyana also has a distinct pre-petroleum low-carbon and forest-finance history that continued alongside expanding oil production. [S01, S04, S07, S22]

The next layer concerns institutional performance rather than chronology. IMF staff documents continuing needs in fiscal anchors, public investment monitoring, procurement, petroleum audit capacity, statistics and public-enterprise oversight. EITI 2026 validation is an ongoing process with publicly recorded stakeholder concerns. Environmental policy includes expanded statutory architecture whose commencement and practical implementation require careful verification. These issues are integral to a useful international case study. [S07, S08, S16, S19]

What an emerging producer can extract from the evidence is a disciplined research method: establish the historical sequence, date each legal change, distinguish entitlement from receipt and expenditure, record ecological hazards at the licence level, publish cross-checkable financial reconciliations, separate climate assets from fossil-fuel accounting, and verify public-service outcomes independently of announcements. Those administrative questions apply across different policy choices. Specific contract terms, saving formulas and energy strategies remain jurisdiction-dependent decisions.

This is a first research manuscript, not a finished monograph. Before final publication the author must complete annual fund reconciliations, contractual and judicial text review, independent environmental and social evidence, a dated 2026 EITI outcome check, quantitative scenarios, comparative-country datasets, external technical review and page-specific reference verification. The proposed 140-160 substantive pages are a future acceptance gate rather than a page count claimed here.

Technical appendix A. Analytical definitions and worked examples

A1: Stock-flow reconciliation. Let F_0 be opening fund cash, D be period petroleum deposits, I be recognized cash investment income, O be other permitted cash flows and W be lawful disbursements. Then $F_1 = F_0 + D + I + O - W$. Disclose transaction timing, currency, investment valuation and whether interest has already been included in D . The Ministry's June 2026 fund balance was USD 4.2942 billion, after USD 1.02 billion in first-half withdrawals. An annual account cannot be assembled by combining that figure with full-year projected deposits without dates. [S07]

A2: Production-to-cash bridge. Petroleum units produced in month M differ from saleable units, cargo lifted in month M and receipts recognized by government in month M . A dated lift ledger should identify cargo, lift date, entitlement owner, grade, price determination, settlement currency, invoice date, deposit date and exchange rate. In H1 2026 the government's 21 lifts included late-recognition effects from 2025. [S07]

A3: Simplified production-sharing example. Normalize a period's gross sales revenue to 100. In the binding 75% ceiling case, assign 75 to illustrative cost recovery and split 25 profit units equally. The state receives 12.5 profit units, plus the separately calculated 2% royalty described by the GRA, giving 14.5 combined units. At a 50% illustrative actual cost share, the split of remaining 50 units yields 25 state profit units plus royalty. At 20%, the corresponding illustration yields 40 plus royalty. These simplified examples omit detailed PSA adjustments and are neither observed 2026 figures nor revenue forecasts. [S02, S15]

A4: Illustrative oil-price sensitivity. A purely hypothetical field selling 1,000,000 barrels per day for 365 days produces 365,000,000 annual barrels. A USD 10 per barrel realized price change moves gross oil sales by USD 3.65 billion before operating expenses, royalty, recoverable costs, fiscal terms, discounting or cash timing. Applying that change directly to the NRF would be incorrect. The case-study model must instead use contract-specific entitlement and fund law.

A5: Capacity versus actual production. Nameplate daily capacity multiplied by 365 describes a theoretical annual maximum before downtime and operational constraints. Actual barrels are the integral of daily measured output, adjusted for reporting boundaries. Government monthly data and independently reported operator annual results should be reconciled.

A6: Carbon versus petroleum cash. A forest-credit agreement describes volume, price, vintage, payment conditions, registry issuance and benefit-sharing obligations. Petroleum cash follows royalty and profit-oil provisions. These may be reported on the same national fiscal page only if separately attributed. Contracted Hess purchase value is not the same thing as realized forest-carbon receipts. [S23]

Technical appendix B. The newcomer decision record

For every proposed instrument, the handbook has eight fields: legal authority; public purpose and measurable output; geological or ecological dependency; evidence required before adoption; implementing institution and budget; independent reviewer; disclosure frequency; and evidence that would trigger revision or suspension. Entries will cover the petroleum licence, environment permit, benefit-sharing instrument, revenue fund, procurement contract, local supplier registry, public investment project and transboundary spill protocol.

Technical appendix C. Questions for external reviewers

- Did the paper apply the legal text in force on the relevant date?
- Did the cost-oil model follow the original executed PSA, including valuation and royalty provisions?
- Did any model count contracted credit sales as cash already received?
- Were unpublished operator assumptions mistakenly treated as regulator-confirmed data?
- Were spill-insurance limits, legal liability and access to immediately liquid compensation distinguished?

- Do Indigenous communities' documented positions appear beyond official summaries?
- Did any poverty estimate use survey data older than the observation year claimed?
- Are state receipts, deposits, parliamentary transfers and spending outputs separately traceable?
- Does the case transfer instruments without recognizing island-state ecological and administrative limits?
- Has an independent expert performed the review, or is a review merely proposed?

References

Full URLs, source types, dates, numerical facts and remaining challenges are recorded in SOURCE_REGISTER.md (S01-S27). Citations inside the final PDF must point to exact pages/sections where available. This edition cites the public original sources by source ID and deliberately does not represent external peer review as completed.

Transferable country handbook

THE NEW-PRODUCER HANDBOOK

A staged institutional workbook for an emerging oil-producing country or island state

Independent companion annex to The Guyana Sequence | Ragunauth Ramsaroop | 27 September 2026

This handbook is a transferability instrument, not a claim that Guyana's history is a universal model. It offers defined questions, document templates and alternative institutional structures from published country cases. Sovereign authorities remain responsible for policy, legislation, fiscal choices, treaty obligations and public consultation. Statements of Guyana's actual history refer to source IDs in SOURCE_REGISTER.md.

1. Start with a resource and institutional map rather than a model law

The first analytical task for a prospective petroleum state is to describe the country before petroleum income changes its financial structure. Guyana's 2009 Low Carbon Development Strategy, earlier commodity economy and decades of offshore exploration predated the first major commercial discovery. The country did not begin building every relevant technical or financial capability only after oil arrived. An island without large tropical forests or comparable offshore acreage has a different starting set of options. [S01, S22]

An initial country profile records offshore geology and the uncertainty of commercial reserves, coastal population and infrastructure, current export composition, public debt, annual tax revenue, national disaster losses, fisheries and tourism dependence, existing environmental agencies, maritime boundaries, regulatory staff, procurement integrity processes, public finance reporting and access to international specialist advice. These are separate datasets. A positive resource discovery changes the geological column but does not erase pre-existing environmental or fiscal exposure. The profile should state the latest observation year for every variable. Missing observations are recorded as missing.

The profile also identifies the institutional decision-makers whose legal mandates may overlap. Typical actors include a parliament, a ministry responsible for resources, finance authorities, a revenue agency, an environmental regulator, a national audit institution, a central bank, a coast guard or maritime authority, regional government, Indigenous or customary-rights institutions, professional regulators and an independent judiciary. Actual institutions differ under civil-law, common-law, hybrid and constitutionally decentralized jurisdictions. The workbook records the current legal mandate rather than silently assigning a new one.

A useful opening deliverable is a two-page public evidence map. The first page identifies the legal source, date of enactment and commencement, administering authority and current

implementing regulations. The second page identifies data held by each institution, reporting lag, legal publication restrictions, unresolved boundary disputes and any independent review process. If existing law does not authorize the proposed activity, the gap is visible before budgets or investment agreements assume implementation.

A separate social and ecological baseline sets the denominator against which later claims will be tested. A coastline with dense tourism and reef ecosystems has a different vulnerability profile from Guyana's low coastal plain and broad forested interior. Climate adaptation, coastal fishery economics, drinking-water resilience and household service levels require direct measurement. Petroleum is a potential revenue source, not a substitute for obtaining the baseline.

2. Exploration stage: six documentary systems

The exploration stage has no guaranteed development or receipts. Guyana's experience includes decades of surveys and unsuccessful wells before Liza. A new state considering costly petroleum-specific institutions should record whether they are scaled to known commercial activity or to uncertain potential. In a country with limited human resources, time-bound specialist contracts and regional sharing agreements are among the institutional forms to compare against a permanent bureaucracy. Neither arrangement removes the need for retained institutional memory or independent conflict-of-interest screening. [S01]

The first documentary system is geological. A searchable block map links the licensing unit, coordinates, bathymetry, historical well records, independent reserve interpretation and uncertainty classification. All petroleum-in-place, contingent-resource and commercially recoverable-reserve statements use explicit definitions. Geological prospects are not parliamentary assets until commerciality is established and sales cash materializes.

The second system is fiscal. Each licence award or proposed agreement has a versioned record of fiscal instruments, stabilization clauses, obligations carried by the contractor, state obligations, allowable costs, relinquishment, abandonment and disputed-area exposure. Where parties negotiate confidentially, the public register states the legal basis and duration of nondisclosure and identifies which post-award terms enter the public domain. Tender dates and the grounds for different procurement methods are traceable.

The third system is ecological. Before drilling, the regulator receives baseline observations of water quality, seabed habitats, fisheries seasons, migration patterns, coastal protected areas, maritime traffic and prevailing currents. For transboundary risks, the record identifies neighboring states and the existing international response agreements. Geological survey and drilling permit terms should not be conflated with consent for a commercial producing field.

The fourth system is technical independence. A national authority can appoint external petroleum engineers, transaction counsel, public accountants and environmental scientists, subject to published conflict controls, confidentiality protections and knowledge-transfer requirements. The record lists each adviser, scope, deliverable, price, tenure, supervision and training obligation. Independent review is not established solely by naming an international consultancy; the published decision file needs evidence of the questions advisers examined and competing options considered.

The fifth system is public participation and rights. A dated register describes legally relevant consultations with fisher groups, coastal settlements, Indigenous and customary-rights holders, tourism representatives and other affected communities. Participation evidence records who received notice, technical materials and translation support, the matters raised and the regulator's reasoned response. National constitutional and treaty obligations determine how consultation, consent, compensation and review operate. The register should not expose personal data or confidential cultural sites.

The sixth system is contingency. A prospective exploration state maps the institutional owner of offshore emergency notification, regional deployment, oil-spill equipment, public-health coordination, aviation and maritime rescue, and access to claims assistance. At this stage a tabletop exercise may demonstrate gaps more clearly than a general emergency-policy statement. Its report identifies response time, jurisdiction and unresolved procurement, not a presumed ability to contain an unmodelled release.

These six systems can be prepared without predicting that commercial oil will be discovered. In an unsuccessful exploration outcome, their data and procedures may still inform geological archives, marine science and public procurement. The financial case for building them at a particular scale remains country-specific.

3. After discovery: the contract and approval decision file

A major commercial discovery introduces a different information problem. The state needs independent evidence about the field's technical and financial viability, while the operator controls substantial reservoir and project-cost information. The period between Guyana's 2015 Liza discovery and December 2019 first oil illustrates why agreements, environmental decisions and fiscal arrangements should be dated separately. Guyana's first NRF law was enacted during this pre-production period. [S01, S02, S04]

The development decision file begins with a technical summary listing verified discovered resources, development phases, forecast decline, required subsea facilities, FPSO or platform capacity, export route, power requirements, drilling schedule and decommissioning concept. A high production forecast must remain labelled as an operator case unless independently confirmed. Where the geology is uncertain, the file records alternative development sequences and their effect on public revenue rather than presenting one central production curve as an inevitable trajectory.

The commercial file compares the actual investment risks allocated by each contract form under the assumptions prevailing when signed. Production-sharing agreements, royalties and taxation, conventional licences and concession-tax systems, service contracts and state equity offer different combinations of upfront funding, price exposure, cost audit, abandonment liability and public oversight. The workbook does not presume any one form is appropriate for a given jurisdiction. It requires internally consistent cash flows, investor-return assumptions and explicit responsibility for failure to reach commercial production.

The fiscal model maintains separate input ledgers for price, recoverable costs, operating costs, investment timing, production, depreciation where relevant, state equity, royalties, profit sharing,

corporate taxation, bonuses and decommissioning. The output ledger distinguishes gross petroleum sales, contractor receipts, the state's profit oil, royalty payments, other taxes, NRF deposits, investment income and money later appropriated through the budget. Each ledger must reconcile on a dated basis.

Guyana's signed 2016 Stabroek agreement includes a 2% royalty, a 75% cost-recovery ceiling and equal sharing of profit oil. Its later model announced a 10% royalty, a 65% recovery ceiling and 10% corporate tax for different licensing circumstances. The older contract's cost bank and the later model should occupy separate scenarios in any comparison. A 2026 cash outcome is not enough to reconstruct the entire risk and bargaining position at the date of the first agreement. [S02, S14, S15]

The environmental approval file is linked to but institutionally distinct from the financial model. A field's commercial value does not establish that mitigation, marine response, compensation, monitoring or financial assurance are adequate. The file identifies each applicable environmental impact assessment, alternatives considered, legal right to inspect and halt operations, monitoring data ownership, baseline methods, spill modelling assumptions and required independent studies.

The public record should preserve dissenting technical views. A regulator's reasoned decision has different evidentiary status from an operator's response to comments, a government announcement or a court ruling. Where litigation is pending, a risk register distinguishes the order in force, the relief requested and legal contingencies still unresolved. Guyana's May 2026 appeal concerning environmental financial assurance demonstrates why final manuscripts must read authenticated judgments instead of relying on competing summaries. [S20]

4. The first-oil readiness rehearsal

First oil often arrives before the initial public development model has stabilized. Under a production-sharing arrangement, cost recovery, crude lifting schedules and sales settlement affect the precise timing of government cash. A national oil accounting system must therefore function during commissioning, not several years later when petroleum receipts already dominate fiscal reporting. Guyana's H1 2026 figures demonstrate the importance of this distinction: some payments received in the first half corresponded to cargoes lifted near the end of 2025. [S07]

The rehearsal opens with a fictitious cargo. The operator supplies a metered production record and government entitlement calculation. The designated fiscal authority checks the volume, crude quality adjustment, eligible costs and royalty; the sales entity checks invoicing and settlement. The central bank confirms receipt. The finance ministry reconciles the money to the NRF statement. The parliamentary record then tracks any authorized transfer from the fund to the consolidated budget. Each step carries a date, accountable official, audit trail and route for a discrepancy.

No example needs to use confidential, live commercial numbers. A deliberately fictional first exercise can contain a missing invoice, late shipment, wrong exchange rate, duplicated cost invoice and incorrect posting of investment income as oil sales. The purpose is to test whether the participating institutions identify and correct differences without an opaque manual adjustment. Material issues found in exercises should appear in a remediation log with a revised protocol and retest date.

At the same time, a parallel environmental drill begins with a hypothetical subsea discharge in poor visibility and a second simulated warning from a neighboring maritime authority. The exercise records the emergency command structure, notification triggers, helicopter and vessel mobilization, sampling, shoreline protection priorities, public communication, legal evidence preservation and claims intake. A state with limited equipment may document formal agreements for shared regional response. The record should distinguish an agreement on paper from a tested mobilization time.

A third rehearsal concerns access to information. Citizens, parliamentarians and researchers try to trace a fictional cargo from the project's regulatory record to the public revenue ledger and verify an appropriation against the published budget. If essential data are lawfully confidential, the portal should specify why and what aggregated fields remain public. A portal with attractive graphics but irreconcilable periods or undefined terms is not a substitute for usable records.

The rehearsals create three different types of evidence: account reconciliation, operating-response capability and public accessibility. None is a measure of the ultimate success of an oil development, but each identifies specific institutional functions before first-year petroleum revenue becomes an ordinary budget assumption.

5. Designing a fund: alternatives rather than a copied withdrawal percentage

Guyana established a statutory Natural Resource Fund before first oil, replaced the first law in 2021 and changed withdrawal provisions in 2024. Ghana's 2011 revenue framework has dedicated stabilization and heritage components. Timor-Leste's Petroleum Fund dates to 2005 with a 2011 amendment. Norway has a mature global fund with a long-run real-return-linked fiscal guideline. The relevant analytical insight is the range of statutory and fiscal architectures, not a ranking of countries or a single numerical withdrawal formula. [S04, S24, S25, S26]

A country examining a resource fund first documents the problem the fund addresses. Annual petroleum cash is exposed to price and production changes. Underground reserves cannot be valued as risk-free cash. Fiscal law often needs a legal method for receiving proceeds, reporting investment performance, transferring a defined amount to the budget and exposing disbursements to ordinary procurement and audit rules. Different development needs, debt service, age structure, exchange-rate arrangements and climate liabilities alter the budget and savings trade-offs.

One analytical model separates stabilization assets from a long-term savings portfolio, with a published transfer rule for each. A second holds a single sovereign portfolio and ties normal-year spending to an estimate of sustainable returns. A third adopts a formula linked to a defined share of verified revenue or assets within constitutional budget procedures. A fourth hybrid model retains a legal spending envelope but adds a multi-year expenditure ceiling, emergency exception and a clear independent disclosure process. Each model creates different accounting, forecasting and governance demands.

The workbook requires a country-specific estimate of the usable cash buffer under a joint commodity-price and natural-disaster stress event. It identifies the unit in which spending obligations arise, the currency and liquidity of fund investments, the risk of simultaneous

domestic recession, and a credible re-entry path after an emergency exception. If a rule permits extra withdrawals during a disaster, the document states who declares the exception, the amount and duration, oversight route and subsequent repayment or recalibration mechanism where applicable.

The core fund reconciliation is ordinary accounting. Opening cash, new petroleum deposits, investment return, other permitted adjustments and withdrawals together explain closing cash. Any transfer into the budget must reconcile to both the fund's statement and the government's budget execution statement. If royalties appear in oil-sale proceeds, they should not be added again under a different financial label. Cash received for prior-year liftings needs a visible timing bridge. One annual press release is not enough to replace account-level controls.

External audits establish whether statements are fairly prepared under an identified accounting standard. They do not establish whether the government selected or delivered the most valuable public projects. That second question requires public investment appraisal, procurement oversight, engineering completion evidence and independent service-delivery evaluation. Guyana's large fund cash balance in June 2026 and the IMF mission's July comments about investment monitoring illustrate the need to keep the two questions separate. [S07, S08]

A useful funding-options dossier therefore contains side-by-side legislation, liquidity simulations, administrative costs, governance responsibilities, spending plans, unknowns and public consultation records. The sovereign authority chooses the design under its constitutional and fiscal system. Publication of alternative assumptions allows later researchers to reconstruct the decision.

6. The petroleum revenue transparency ledger

An extractive revenue website serves different audiences. Parliament needs statutory compliance and account reconciliations. A central bank needs cash management and portfolio controls. Tax authorities need expenditure verification and settlement records. Local communities need understandable information about revenues, distribution programmes and environmental incidents. Researchers need data definitions and historical versions. A public ledger therefore needs more than a current total.

The minimal petroleum portion lists the project and contract identifier, fiscal instrument, observation period, quantity and unit, price basis, lift date, government entitlement, invoice period, payment date, receiving account and audit status. Project-specific records with legally protected commercial information may be aggregated, but the scope and legal basis of aggregation should be documented. An entity-level register records beneficial-ownership disclosures and changes subject to applicable privacy and security law.

The fund portion lists opening assets, new deposits by category, investment interest or income, valuation adjustment, lawful budget transfers and ending cash. Each published table includes its accounting basis and revision history. The budget portion separately identifies appropriations, procurement contract values, actual spending, verified physical completion, maintenance allocations and beneficiary results. A reader should be able to follow cash without assuming every transfer immediately produces a completed asset.

Environmental transparency is part of the same programme but has a different standard of evidence. The public permit register identifies drilling and production authorizations, impact assessment dates, permit expiry, reportable incident definitions, measured emissions, significant enforcement decisions, available monitoring data and any material remediation obligations. The administrative record captures which regulator confirmed a fact and which claim rests on operator self-report.

Civil-society representation cannot be reduced to web access. The EITI Board's May 2026 decision recorded stakeholder concerns about civil-society selection and initiated a full validation with external participation. An implementation workbook therefore documents who selected representatives, applicable rules, whether affected groups had access to technical support, how dissenting submissions were considered and whether public records provide a traceable response. The 2026 process should be described according to its final verified status before publication. [S16, S17]

A release process protects analytical integrity. Each dataset includes version, publication date, source agency, applicable period, methodological changes and a clear correction notice. A monthly series that changes definitions halfway through the year should not appear as a continuous trend without reconciliation. Differences between production, exports, oil sales and deposits should be shown rather than obscured.

7. Local participation and the value-added question

Governments use several forms of local participation: supplier qualification, procurement preferences or reservations, mandatory reporting, technical training, joint ventures, domestic processing, public education and access to finance. The domestic economic outcome depends on the mix of actual productive inputs rather than one gross procurement figure. Guyana's 2021 Local Content Act and its 2026 procurement reporting offer a case for examining the difference between an administratively eligible supplier and independently measured domestic value added. [S05, S07]

The workbook separates business registration, beneficial ownership, share of labour paid locally, imports embedded in purchased goods, plant and equipment owned within the jurisdiction, local subcontracting, professional accreditation, technology transfer and taxes retained. A service supplier may be a meaningful domestic employer even with imported equipment. Another company may book large contract value through a local subsidiary with most technical work completed abroad. Without disaggregation, both transactions look similar in gross purchase statistics.

A registry should permit independently checkable ownership and skills claims while respecting commercial confidentiality and individual privacy. Independent verification might compare procurement statements with audited supplier accounts, customs classifications, professional training completion and aggregate labour statistics. Alternative explanations for spending increases, including domestic construction inflation or reclassification of supplier categories, belong in the same analytical report.

Geography is equally important. Offshore supply chains often concentrate near ports and the capital, while inland and remote districts face different infrastructure needs. A national local-content percentage says little about benefits to hinterland communities. A map of procurement distribution, pay and completed infrastructure is therefore more informative than a single national total. For a small island, limited land, port throughput, environmental carrying capacity and seasonal tourism affect local supply opportunities.

The analysis must also measure possible displacement. Large operator contracts may raise wage rates and absorb engineers, drivers and electricians previously employed by clinics, schools, municipal agencies and non-oil firms. The relevant question is whether new training and productive capacity expand the overall skilled workforce faster than demand, not merely whether the petroleum industry hires more people. Labour-force and wage data require source, sample period and inflation adjustment.

8. Indigenous peoples, coastal livelihoods and public participation

An ethical and analytical case study makes participation visible as a process and an outcome. Guyana's forest-carbon policy commits part of eligible proceeds to Indigenous village-led development plans. Offshore petroleum activities primarily expose marine and coastal interests, while national spending decisions affect interior communities. Both require independently documented benefits and grievances. A small island's customary marine tenure, coastal fishery governance and tourism land rights may present different institutions and legal obligations. [S22, S23]

The country profile maps statutory land tenure, customary resource rights, relevant constitutional provisions, applicable international obligations, agreed consultation procedures and environmental justice claims. The appropriate consultation and consent process depends on those instruments and the circumstances of the project. Evidence includes dates, disclosure formats, translations, support for independent technical review and a record of the response to documented objections. A signed attendance sheet alone does not establish informed participation.

Community-finance claims require four distinct records: funds legally designated, sums transferred, locally selected projects and independently observed outcomes. A documented transfer to a village account is one fact. Community consent to a project, completion of the project, continuing operation and resulting improvements are additional facts requiring their own evidence. The full white paper should preserve contrary testimony and avoid extrapolating from a handful of cases to every participating village.

Maritime impacts require similar attention. Fishers' exposure depends on where vessels operate, exclusion zones, changes in landings, port congestion, prices, compensation procedures and baseline variability. An increase in national GDP does not prove unaffected fishers benefited. A properly specified marine impact study accounts for weather, seasonal migration, fisheries management, other industries and changes in reporting. The state and operator's ecological claims need independent verification where data are available.

The publication protocol should provide a right of factual response for materially criticized institutions and stakeholder organizations without allowing any source to veto findings. It should protect respondents who face credible confidentiality or retaliation risks. A clear chronology of complaints, agency responses and court decisions is more useful than a general assertion of agreement.

9. Environmental assurance across the full project lifecycle

Environmental permitting and oil-spill financial assurance are related but distinct. Licensing decides whether and under what conditions exploration or production is lawful. Financial assurance addresses availability and enforceability of resources after a covered event. Emergency response identifies assets and decisions available within the first hours and days. Ecological restoration addresses losses long after the emergency ends. Confusing these categories can create false confidence or understate actual liabilities.

Guyana's 2025 oil-pollution legislation and May 2026 appeal over permit-linked financial assurances offer detailed subjects for legal verification. The 2026 mid-year report described implementing regulations as under development and commencement as a future event. At a new publication cut-off, enacted legislation, operational regulations, permit guarantees and appellate orders must each be verified separately. This handbook does not assume any quoted insurance total without the actual contractual documents. [S07, S19, S20]

A project-specific risk register records the hazard, expected pathways, sensitive receptors, probability model and its uncertainty, monitoring threshold, legal reporting trigger, escalation owner, containment strategy, finance instrument and external response partner. Where the country lacks credible dispersion modelling or ocean-current data, the register states the limitation instead of displaying unwarranted decimal precision. Regional assistance agreements require contact details, transit permissions, operational compatibility and periodic exercises.

Insurance due diligence examines covered activities, limit per occurrence and aggregate, exclusions, deductible, insurer creditworthiness, jurisdiction for claims, financial assurance beneficiary, rapid-access liquidity, environmental restoration scope, third-party claims and decommissioning. A parent-company guarantee raises separate questions about capacity, jurisdiction, enforceability and its interaction with an operator's continuing liability. Evidence of a guarantee's existence is different from evidence that public liabilities are fully secured.

Monitoring covers normal operations as well as exceptional spills. Discharge records, produced-water treatment, routine and non-routine flaring, methane measurement, drilling waste, ballast handling, vessel collisions, protected species, beach contamination, accidental chemical release and supply-base impacts need project-specific methods and reporting thresholds. Annual mean conditions should not conceal short high-intensity incidents.

A small-island adaptation module places reef health, mangroves, beaches, freshwater lenses, tourism seasonality and evacuation logistics into the response plan. Guyana's forest MRV and coastal flood adaptation experience contain relevant monitoring methods, but the appropriate ecological baseline differs. A transferability assessment states precisely which laboratory

methods, satellite products, emergency drills or public disclosure practices travel between jurisdictions and which require new expertise.

10. Public investment and durable service outcomes

Converting exhaustible revenue into durable national assets is a broader question than saving revenue in an investment fund. Infrastructure and human capital have different delivery schedules and lifecycle liabilities. Large petroleum receipts create an opportunity to finance capital projects but also expose a small economy to construction cost inflation, workforce shortages, poor maintenance planning and irreversible land-use changes. IMF staff's July 2026 mission highlighted Guyana's need for sound public investment appraisal and monitoring. [S08]

The public investment record begins with a published definition of the service problem, baseline conditions, alternatives, demand assumptions, environmental costs, construction schedule and lifecycle operation. For a hospital, the completion test includes staffing, power, maintenance, service hours and patient outcomes, not merely the ceremonial opening. For roads, the test covers load conditions, all-weather access, drainage and maintenance. For coastal barriers, the test covers projected flood risk, residual exposure and local ecological impacts.

The procurement register connects each project to tender design, qualified bidders where publicly available, evaluation criteria, contract award, change orders, schedule extensions, independent inspection and defects liability. It distinguishes price savings during tender from total cost after scope changes. A public project portfolio report should identify projects whose maintenance liabilities exceed planned operating budgets before those assets enter service.

Fiscal analysis compares infrastructure delivery capacity with projected annual petroleum cash under multiple prices and production paths. A legally available withdrawal is not evidence that domestic construction supply can absorb the resulting spending without delays or price escalation. A separate model estimates import demand, skilled labour, land requirements and potential crowding of non-oil investment. Uncertainty in domestic absorption is reported alongside uncertainty in the oil price.

Social outcomes need disaggregation. Regional living standards, access to water and electricity, education completion, maternal and child health, housing affordability, travel time and employment among women and younger workers are plausible indicators. Their applicability and target levels follow national development plans and community consultation. Surveys taken years before first oil should remain marked historical rather than be presented as current evidence.

11. Five-year observation schedule

The workbook groups observation by operational stage rather than guaranteeing a calendar date for oil production. Year 1 begins when a jurisdiction formally launches a resource-governance work programme, whether or not a commercial discovery exists. Every item below is a record requested for research and possible administrative review, not a mandatory legal prescription.

During months 0-3 the record covers petroleum licence and geological holdings, current law and commencement status, fiscal exposures, marine baseline, existing disaster protocols and the capacity of existing agencies. The finance ministry assembles a preliminary risk and revenue-uncertainty register. A ministry or regulator charged with petroleum maintains the licensing and contract inventory. Statutory audit bodies map their access to records. Representative stakeholder institutions identify consultation requirements under applicable law.

During months 4-6 the record includes independent technical advice procured, review of existing disclosure obligations, available fiscal and contract design alternatives, cost-audit capacity, ecological knowledge gaps and an initial incident communications exercise. Where petroleum prospects are not commercially proven, the budget scenario uses zero commercial oil revenue as its central binding reality rather than projected receipts. If a commercial discovery has occurred, engineering and fiscal scenarios identify range and timing.

During months 7-12 the country dossier details chosen regulatory and contract procedures, EIA scoping, transboundary notifications where required, future fund legislation options, sample reconciliation records, independent audit mandates and a funded skills plan. Publication may be staged in line with lawful data protection, but the record explains disclosure omissions. A tabletop exercise tests resource and environmental decisions across agencies with no assumptions of operational perfection.

During months 13-24 the observed documents reflect the actual development stage: environmental permit record, drilling or commissioning inspections, contract-cost audits, any effective resource fund legislation, validated production metering where applicable and reporting systems. A jurisdiction still in exploration does not need fictitious profit-oil tables; it records progress against exploration-stage tasks instead. Any delay in production updates the fiscal scenario and budget expectations.

During years 3-5 the portfolio review examines signed versus actual development schedules, year-by-year revenue reconciliations if production has begun, actual fund deposits and transfers, fiscal stress-test assumptions, public investment delivery, domestic-value measurements, environmental incidents, administrative appeal outcomes and independent community outcome evidence. Comparable historical series require a fixed base and disclose methodological revisions.

At each stage the workbook reserves a correction section. A delayed field, adverse environmental judgment, commodity shock or public procurement failure changes facts, which changes the scenario. Reviewers should be able to reconstruct the original assumptions and the date those assumptions were revised. The workbook does not erase an earlier forecast when a later publication supersedes it.

12. Ten cross-agency exercises for the implementation team

Exercise one asks the petroleum ministry and environmental regulator to trace a fictional exploration block from acreage designation through EIA scope and final approval. Record the public legal documents and any unresolved overlap of powers.

Exercise two asks independent technical counsel to quantify how a dry-well outcome changes the proposed administrative budget. Document the difference between spending on general-purpose geological data and spending that depends on an operating hydrocarbon industry.

Exercise three uses the executed PSA or concession terms to trace a fictional field's sales, allowable cost, royalty, taxation and state receipt. The revenue authority records how disputes and payment delays affect cash timing.

Exercise four asks the central bank, finance ministry and auditor to reconcile the same fictional oil payment across accounts, including a settlement posted one month after physical lifting. A mismatch creates a correction record, not a hidden balancing item.

Exercise five asks the parliamentary budget office to distinguish legally available fund transfers from cash already collected, projected year-end receipts and operational spending commitments. Run an illustrative low-price year and document different buffer designs without prescribing a single fiscal rule.

Exercise six asks the environmental regulator and coast guard to test a simulated spill with a neighboring state affected. Record who authorizes sampling, emergency procurement, shoreline access and claims intake. Repeat the same exercise under a power outage or severe coastal storm.

Exercise seven asks the procurement authority to evaluate a large petroleum-financed infrastructure project before and after a hypothetical 25% construction-cost overrun. Include maintenance, skill shortages and the consequences of delaying other projects.

Exercise eight asks the local supplier registry and an independent economist to reconcile a high gross procurement announcement with wage payments, imported equipment, local ownership, subcontracting and retained domestic profits.

Exercise nine asks a protected independent panel to document stakeholder participation in a fictional coastal development approval. The test is whether different views reach the decision-maker and receive reasoned responses, not whether every party agrees.

Exercise ten asks a mixed technical review team to assess a hypothetical halt in offshore output coinciding with flooding and a sharp oil-price fall. Recalculate cash, essential services, contract obligations, ecological emergency funding and the credibility of public communications.

Every exercise produces a date, participating institution, source documents, observations, corrective actions, verification method and retest date. An uncompleted exercise is shown as outstanding rather than assigned a favorable qualitative label.

13. What Guyana's case does not establish

Guyana's early NRF legislation is a historical observation. It does not by itself prove that a particular savings rate, withdrawal formula or investment portfolio has maximized intergenerational welfare. Such a claim would need long-run counterfactual analysis, accounting and household outcomes. A large fund balance and rapid production growth are not interchangeable with measured school performance, flood resilience or reductions in poverty. [S04, S07, S10]

The 2016 and later model PSAs document different fiscal terms in different commercial circumstances. These documents do not establish that the same contract works for a shallow-water field, a field with state equity, a landlocked onshore producer or an island whose public balance sheet cannot bear a large contingent spill liability. The exact risk allocation and counterparty financing capacity must be reconstructed for each case. [S02, S14]

Guyana's forest programme offers a separate climate-finance case, but a country without forests of comparable scale has no identical carbon asset. A country's petroleum revenues and forest-credit receipts have different legal rights, accounting bases and environmental effects. Carbon-credit eligibility and contract value do not establish measured nationwide carbon neutrality or a permanent right to petroleum expansion. [S22, S23]

The existence of enacted pollution legislation is not synonymous with commencement, funded enforcement or tested emergency operations. An appeal about financial assurance does not settle every possible ecological-loss scenario. The research must retain authenticated legal sources and independent insurance analysis as open requirements. [S19, S20]

Finally, country comparisons have different production histories and social baselines. Ghana's revenue law, Norway's mature financial fund, Timor-Leste's sovereign petroleum fund and Suriname's pre-first-oil preparations are documentary comparators rather than a league table. A useful successor study would collect consistent country-year data, label changes in accounting methodology, document the distinct welfare measures available and subject any causal claims to a defensible identification strategy. [S24-S27]

14. Compact reusable templates

Country opening record: Name of jurisdiction; programme initiation date; discovery status; resource classification; public debt observation year; existing royalty and tax base; existing environmental and marine protections; sensitive ecosystems; customary-rights framework; coastal and inland settlements; responsible institutions; independent audit mandate; unresolved treaty and maritime questions; last household-survey observation year; central-bank data accessibility.

Agreement comparison record: Contract identifier and version; signature and effective dates; contract model; legal approval; resource and price uncertainty at the time; development and abandonment obligations; royalty, tax and profit-sharing terms; cost eligibility and audit rights; stabilization and dispute resolution; assignment and beneficial ownership; environmental and financial assurance; public disclosure status; independent counsel and conflict review.

Fund reconciliation record: Fiscal period; audit status; opening cash and investment portfolio value; petroleum deposits by source; investment income and valuation adjustments; authorized withdrawal; receiving treasury account; budget-law reference; closing cash; reconciliation difference; publication and revision dates; public audit opinion.

Environmental assurance record: Project and licence identifier; baseline surveys; EIA date; permit conditions; discharge and emissions thresholds; incident reports; independent monitoring; response assets and mobilization results; insurance contract and exclusions verified; parent or other guarantee where applicable; claim jurisdiction; decommissioning security; statutory effective dates; public grievance and judicial review procedures.

Community outcomes record: Affected stakeholder class; legal consultation rule; disclosure and meeting dates; dissenting and supporting submissions; reasoned agency response; monetary entitlement and transfer; locally approved projects; construction and operation verification; measured benefit and distribution; continuing grievances; independent reviewer.

Public investment outcome record: Project identifier; approved business case; procurement method; financing mix; cost baseline; contract award and variations; completion evidence; maintenance provision; climate or environmental exposure; service baseline; actual beneficiaries; operation performance; auditor's findings; outstanding remedial work.

Correction and disclosure record: Dataset or legal assertion corrected; original publication date; conflicting source; confidence and reason for correction; old and new values; whether previous analytical conclusions change; approving researcher; next source verification date; cross-referenced exhibits.

The final 140-160 substantive-page monograph will integrate these templates with year-by-year datasets, a verified comparison of fund rules, legal text citations, multi-scenario modelling and independent external technical review. Until then, this playbook is an openly labelled research instrument attached to the working edition.

Source register

Dated source IDs correspond to citations in the manuscript. Public URLs link to originals. Key unresolved verification tasks appear after the register.

S01 ExxonMobil Guyana 2023 Annual Report

Early seismic and onshore exploration, May 2015 Liza commercial discovery, 2019 first oil and FPSO dates

Limit: Operator is an interested party; cross-check public licence register

<https://corporate.exxonmobil.com/locations/guyana/publications/2023-annual-report>

S02 Guyana Ministry of Natural Resources, Stabroek petroleum agreement

Official text of the 2016 agreement, contract terms and baseline obligations

Limit: Distinct from future PSAs and public summaries

<https://petroleum.gov.gy/docs-and-pubs/petroleum-agreement-stabroek-block/>

S03 Ministry of Natural Resources, block register

June 27 2016 agreement / October 7 2016 effective date and licence information

Limit: Different contract milestone dates refer to different acts

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S04 2024 NRF Annual Report, Ministry of Finance, notes/legal framework

NRF law assented January 23 2019; replaced December 30 2021; withdrawal rule amended February 6 2024

Limit: Full audited report must be checked for changing cash period

https://finance.gov.gy/wp-content/uploads/2025/11/2024_NRF_Annual_Report_Final_Draft-JD0715_GE1507-Final-Print.pdf

S05 Parliament of Guyana, Acts index

Local Content Act 2021 and NRF Act 2021 parliamentary dates

Limit: Parliamentary passing, assenting and commencement dates differ

<https://www.parliament.gov.gy/publications/acts-of-parliament/P80>

S06 Finance, 2026 Budget Speech, January 26 2026, 124-page PDF

January production/revenue/NRF assumptions and policy descriptions

Limit: January projections were superseded by September mid-year update

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S07 Finance, 2026 Mid-Year Report, issued September 2026, printed pp. 21-25 and 104

H1 2026 21 government lifts, USD 1,778.6m profit-oil proceeds, USD 218.4m royalties, USD 4,294.2m end-June fund cash, USD 1,020m withdrawals; revised forecast USD 6,497.6m petroleum deposits

Limit: Government estimates and projections, not audited end-year results

<https://finance.gov.gy/wp-content/uploads/2026/09/Mid-Year-Report-2026.pdf>

S08 IMF, July 31 2026 mission concluding statement

IMF staff views on balance/withdrawal rule, absorption, audit, procurement, AML/CFT, statistical gaps and LCDS

Limit: Staff preliminary findings, not IMF Executive Board Article IV report

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S09 IMF, 2025 Article IV report and press

Detailed macro, fiscal and debt context, external baseline

Limit: Historical to 2024; do not use as September 2026 snapshot

<https://www.imf.org/en/publications/cr/issues/2025/05/07/guyana-2025-article-iv-consultation-press-release-staff-report-and-statement-by-the-566712>

S10 World Bank, Guyana country overview

Social indicators, non-oil growth and lack of current household poverty data

Limit: Explicitly labels last cited poverty estimate 2019, not 2026

<https://www.worldbank.org/ext/en/country/guyana>

S11 ExxonMobil, Yellowtail startup, August 8 2025

Fourth FPSO, installed capacity exceeded 900k bopd

Limit: Installed capacity is not sustained actual output

https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project

S12 ExxonMobil, November 12 2025

900k bopd daily output milestone

Limit: A single-day threshold is not 2025 annual average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

S13 Petroleum Management Programme oil-production chart

Government monthly production chart through July 31 2026

Limit: Preserve date and distinguish unit from capacity

<https://petroleum.gov.gy/data-chart/oil-production/>

S14 Government of Guyana, 2023 model fiscal terms

Announced 10% royalty, 65% cost ceiling and 10% corporate tax for future licensing

Limit: Do not apply revised terms retroactively to Stabroek

<https://petroleum.gov.gy/article/govt-announces-wave-of-enhanced-fiscal-terms-for-first-offshore-oil-and-gas-competitive-licensing-round-highlights/>

S15 Guyana Revenue Authority, royalty treatment, 2026

2016 2% royalty added to simplified 12.5% government profit share when cap binds

Limit: 14.5% example is not constant government take across the life of the block

<https://gra.gov.gy/royalty-payments-to-guyana-government/>

S16 EITI, May 12 2026 Board decision

Second validation from May 15; concern over civil-society selection and stakeholder confidence

Limit: No 2026 final validation result has been established in this work

<https://eiti.org/es/node/28602>

S17 EITI, 2026 call for stakeholder views

2022 validation and 2026 validation scope

Limit: Distinguish earlier assessment from later validation

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S18 Guyana EITI country secretariat

December 2025 report covers fiscal year 2023; validation visits June 2026

Limit: Secretariat claims require triangulation with EITI global board

<https://eiti.gy/about/>

S19 Parliament, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Enacted Act No. 6 of 2025

Limit: September mid-year report says draft regulations progressing and commencement expected end-2026; enactment is not commencement

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S20 Government statement about May 7 2026 Court of Appeal decision

Appeal set aside High Court unlimited financial-guarantee order

Limit: Obtain authenticated judgment and independent legal commentary before final jurisprudential analysis

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S21 EPA environmental authorisation services

Environmental authorisation/permit process and regulated-project categories

Limit: Online application portal alone does not prove monitoring quality

<https://eservices.epa.gov.gy/permits>

S22 Guyana LCDS 2030, October 2024 update

2009 framework, Norway USD 227m for 2009-15, 2022 policy and forest-credit programme; 15% community allocation

Limit: LCDS describes government strategy and self-reported expenditure; impact audit separate

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S23 LCDS, Guyana ART TREES fact sheet, 2022

Hess minimum USD 750m purchase agreement, credit volume and minimum prices

Limit: Contracted value is not money already received

<https://lcds.gov.gy/fact-sheet-on-guyanas-engagement-in-art-trees/>

S24 Ghana Petroleum Revenue Management Act 2011

Stabilisation Fund, Heritage Fund, publication cadence and public accountability

Limit: Comparator institutional design, not a performance ranking

<https://ghalii.org/akn/gh/act/2011/815/eng%402011-04-15>

S25 Norway, 2025 Government Pension Fund white paper

Entire net state petroleum cash transferred to GPF; 3% real-return fiscal guideline

Limit: Norway's long accumulation and mature state capacity differ from a new producer

<https://www.regjeringen.no/contentassets/185dca55ef5649d1bc08db2699bb861f/en-gb/pdfs/stm202420250022000engpdfs.pdf>

S26 Timor-Leste central bank, 2005/2011 Petroleum Fund statutes

Statutory fund structure and amendments

Limit: Cross-check 2026 depletion pressures independently before claims about outcomes

https://www.bancocentral.tl/publication/laws_np/

S27 TotalEnergies Suriname, September 19 2026

GranMorgu 2024 FID, first oil planned 2028

Limit: Planned start is not operating output

<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>

S28 Guyana pre-oil sovereign wealth fund Green Paper

Documents work beginning in late 2016 with Commonwealth Secretariat support, inter-agency working group in 2017, IMF missions and FARI training before first oil

Limit: Historical government policy paper. Treat forecasts in the paper as superseded by actual later events

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Green-Paper-Final.pdf>

S29 IMF 2023 Article IV staff report

Explains two-rule system: Stabroek PSA revenue sharing plus NRF withdrawal framework; notes first oil receipts deposited in 2020 and first budget transfer in 2022; records 2023 model-PSA changes and governance priorities

Limit: IMF analysis reflects information available in 2023 and should not substitute for 2026 data

<https://www.elibrary.imf.org/view/journals/002/2023/379/article-A001-en.xml>

S30 Natural Resource Governance Institute, natural-resource fund governance guidance

Clear objectives, deposit/withdrawal rules, investment rules, independent oversight, broad consensus and budget integration; warns a fund alone does not produce good governance

Limit: Comparative policy guidance, not jurisdiction-specific legal advice

https://resourcegovernance.org/sites/default/files/NRF_Complete_Report_EN.pdf

S31 NRGi sovereign wealth fund guidance for policymakers

Highlights oversight, disclosure, fiscal-rule fit, dangers of direct domestic investment from funds and the need

to fit rules to national conditions

Limit: Older cross-country guidance; current country practices require fresh verification

<https://resourcegovernance.org/articles/sovereign-wealth-funds-guidance-policymakers>

S32 LCDS homepage, accessed September 2026

Current official payment dashboard lists cumulative Phase II carbon-credit payments of USD 353.5m, including March 2026 USD 57.9m

Limit: Official programme reporting; independent audit of final use and outcomes remains separate

<https://lcds.gov.gy/>

S33 EITI validation schedule, accessed September 2026

Guyana validation commenced 15 May 2026 and remains listed among ongoing validations

Limit: Do not state a final 2026 validation score until EITI publishes a decision

<https://eiti.org/validation-schedule>

S34 EITI Board decision 2026-19, 12 May 2026

Full validation launched with external expert; Board expressed concern over continuing lack of full stakeholder confidence in civil-society representative selection

Limit: Board concern is attributed to EITI, not adopted as the author's independent conclusion

<https://eiti.org/es/node/28602>

Pending verification

Priority unresolved checks before final publication

1. Obtain authenticated PDF text of the May 2026 Court of Appeal judgment and any later appeal.
2. Verify whether the 2025 spill act was commenced after the June 2026 reporting cut-off and before 27 September 2026.
3. Reconcile public Bank of Guyana fund statements to Ministry of Finance June 2026 balance, petroleum deposit and interest categories.
4. Determine public status of 2018-2020 and later cost-oil audits and any contested arbitration.
5. Obtain original EPA permit schedules, environmental impact assessments, baseline marine datasets, third-party incident logs and the precise insurance/parent guarantee instruments.
6. Verify 2026 EITI validation report status at press time, not solely government statements.
7. Independent assessment of Indigenous consultation and village-level benefit delivery with free, prior and informed consultation standards relevant to the jurisdiction.
8. Distinguish project expenditure from commissioned/working infrastructure and verify costs, maintenance and beneficiaries.
9. Validate local content supplier ownership and incremental local value added independently of gross spend.
10. Conduct external technical, financial, legal and environmental review before calling this paper peer reviewed.